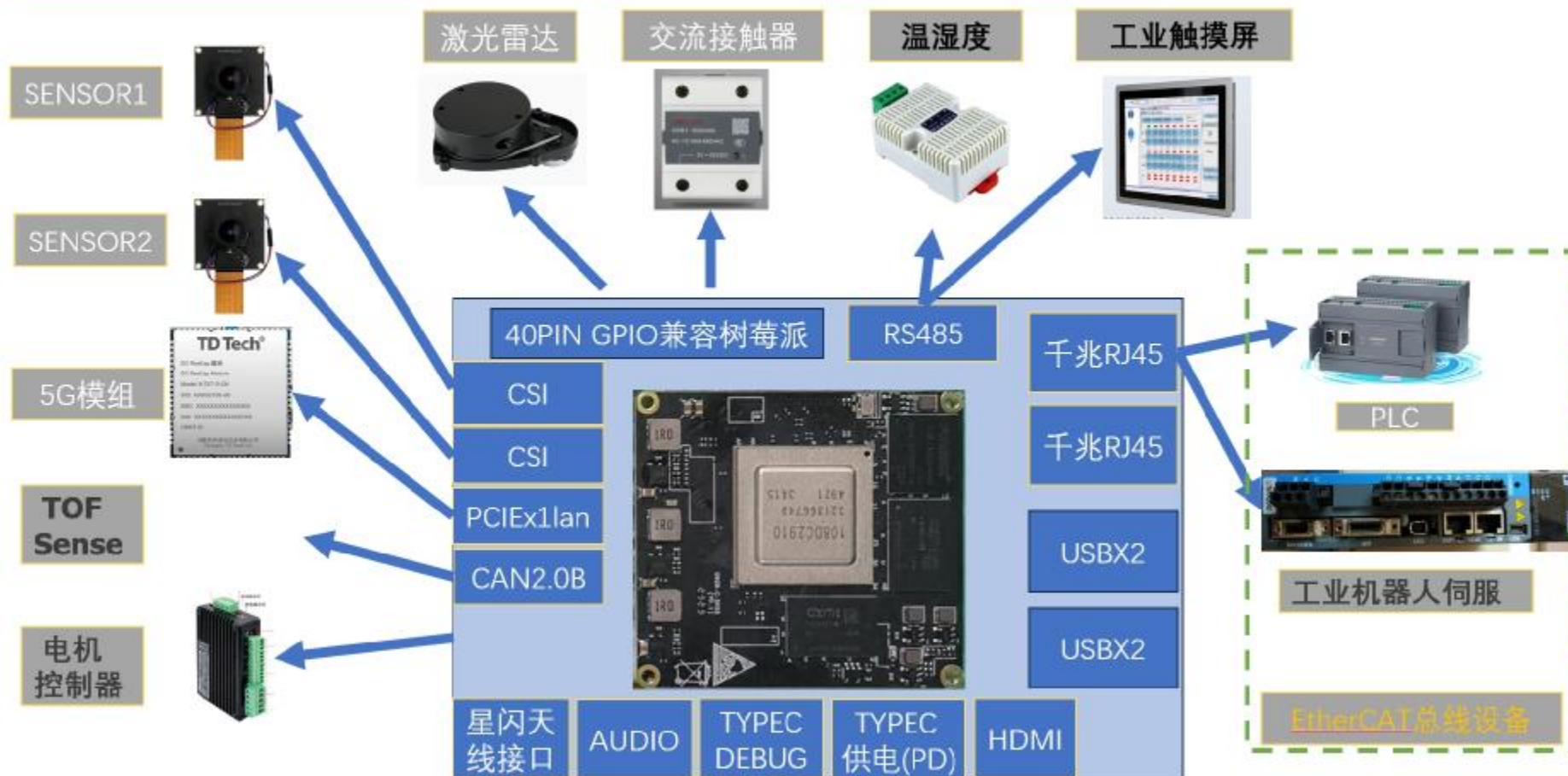
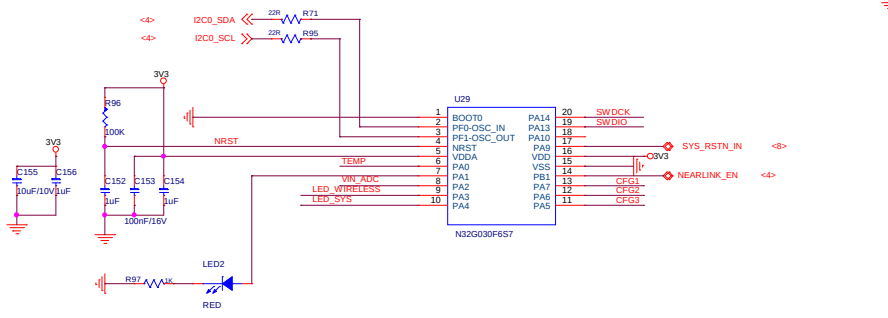
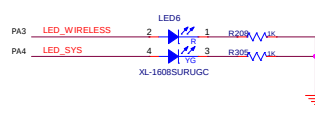
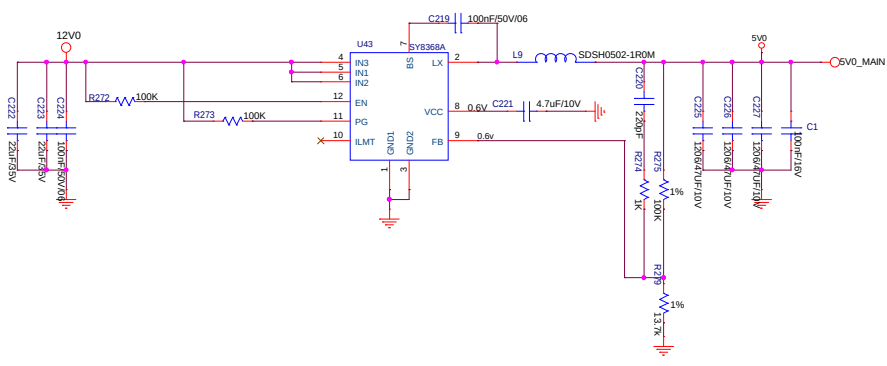
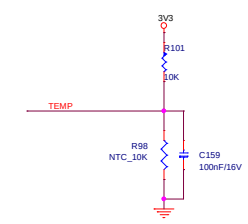
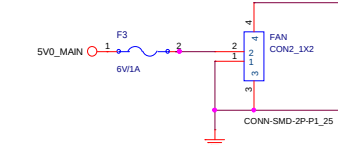
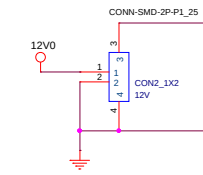
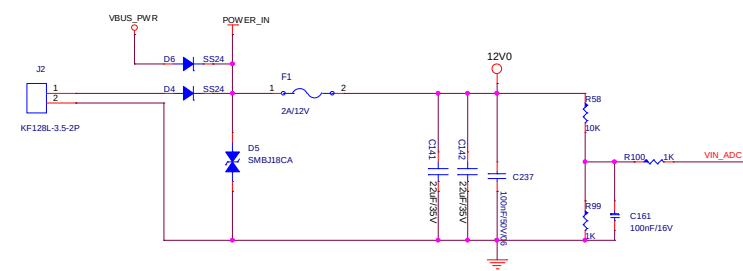
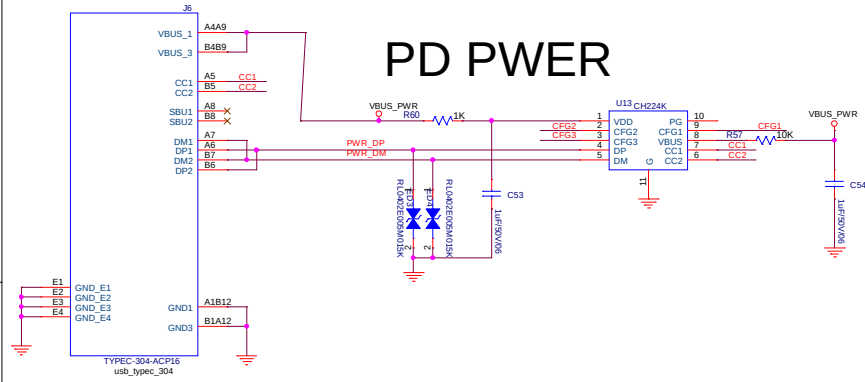


欧拉排外设规划，核心板+底板

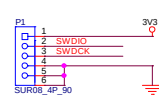
Ebaina®

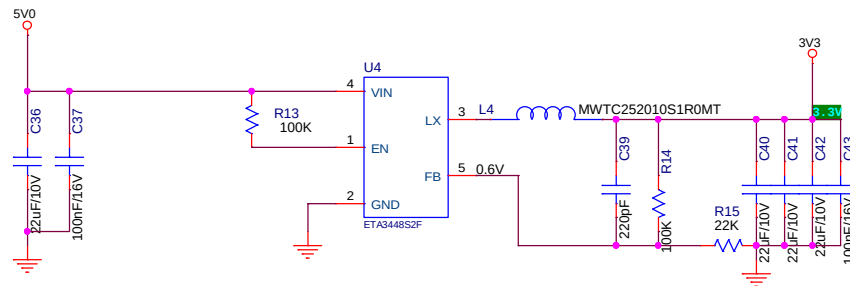


PD PWER

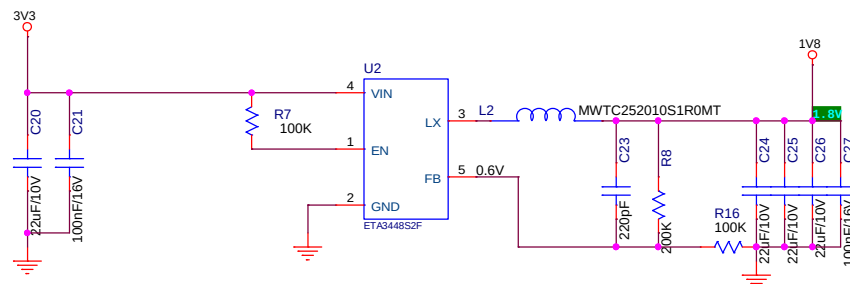


指示灯优化(复位和升级按钮):
 A. 电源指示灯从电源MCU引出, 单色红色LED灯
 红色常亮: 协商成功, 正常供电
 红色闪烁: 协商中(80%左右时间)
 红色熄灭: 协商失败
 B. 50mW绿色指示灯变为绿色和红色双色指示灯,
 并从电源MCU引出
 绿色常亮: 代表系统正常启动
 绿色熄灭: 代表系统启动中
 红色常亮: 系统故障1: wIFI和蓝牙模块异常
 红色熄灭: 系统故障2: 上下位机通信异常
 绿色常亮+红色: 系统启动正常, 但没有和云端服务器连接
 绿色熄灭+红色: 50mW启动正常, 但无法上网(未插SIM卡)
 2. 取消12V供电插座旁边的电源协商指示灯。

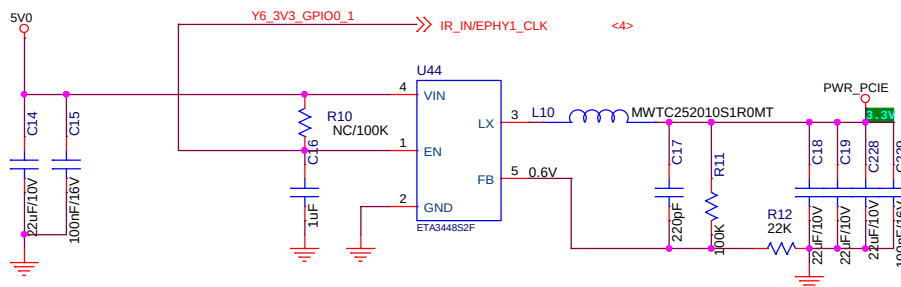




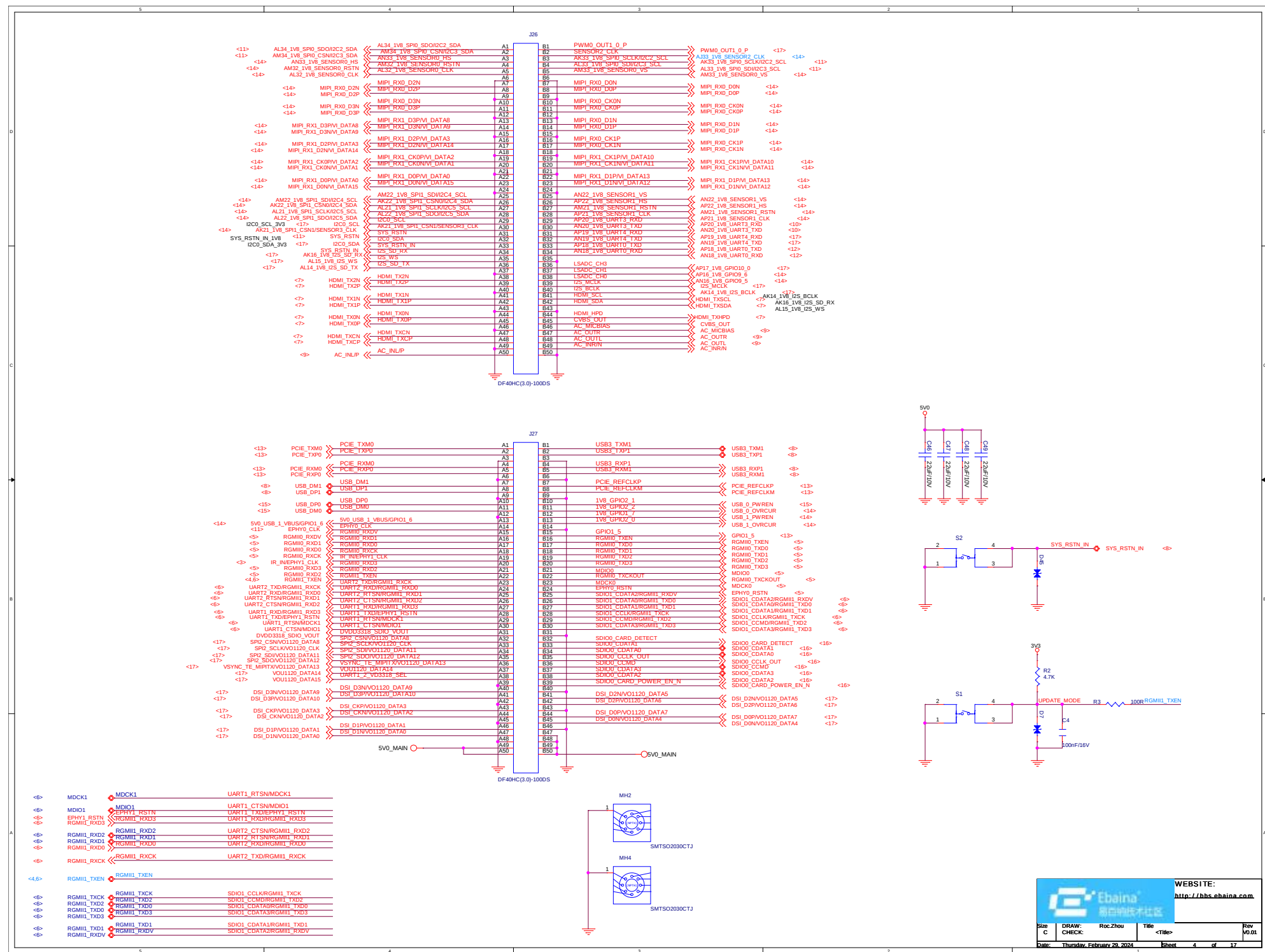
$$V_{out} = 0.6 * (1 + 100k/22k) = 3V3$$



$$V_{out} = 0.6 * (1 + 200k/100k) = 1V8$$



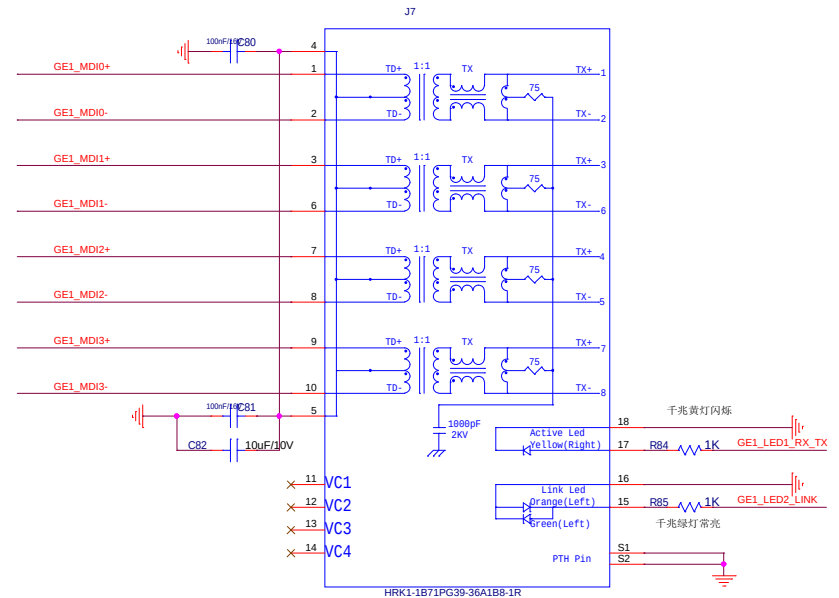
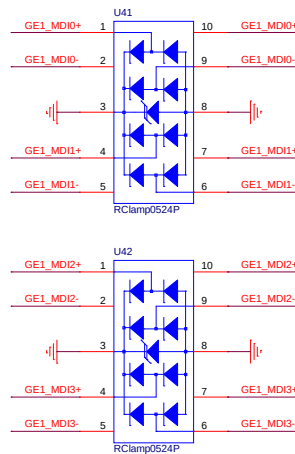
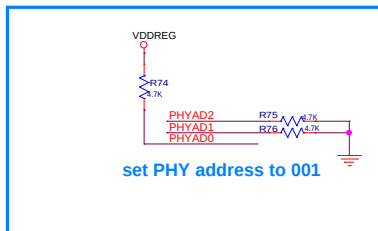
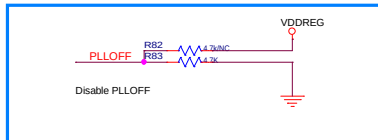
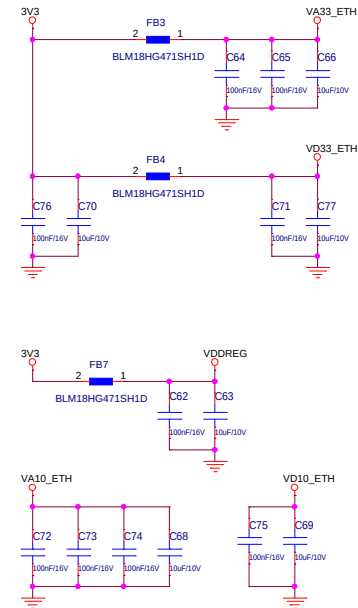
$$V_{out} = 0.6 * (1 + 100k/22.1k) = 3V3$$



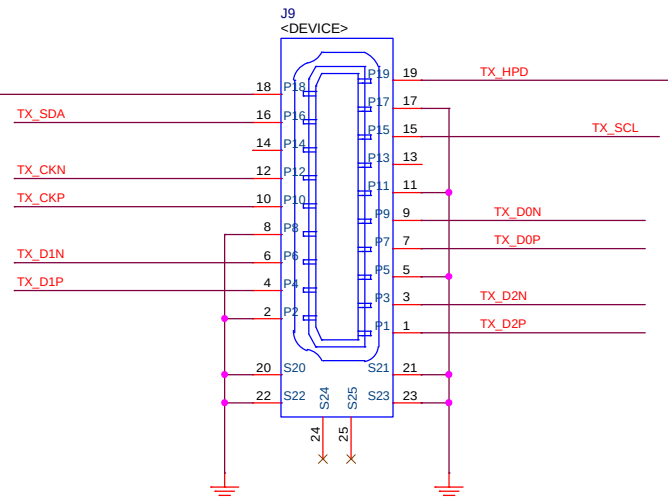
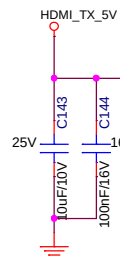
The schematic diagram illustrates the connection between the STM32F405VGT6 microcontroller and the RTL8211F-CG Ethernet PHY. The microcontroller's pins are connected to the PHY's pins as follows:

- Power Supply:**
 - VDDIO (Pin 11) is connected to VDDIO_ETH (Pin 3).
 - VDDREG (Pin 28) is connected to VDDREG_ETH (Pin 30).
 - MDCK1 (Pin 13) is connected to MDCK1 (Pin 13).
 - MDIO1 (Pin 14) is connected to MDIO1 (Pin 14).
- Control Signals:**
 - TXD3 (Pin 15) is connected to TXD3 (Pin 15).
 - TXD1 (Pin 16) is connected to TXD1 (Pin 16).
 - TXD0 (Pin 17) is connected to TXD0 (Pin 17).
 - TXEN (Pin 18) is connected to TXEN (Pin 18).
 - TXCK (Pin 19) is connected to TXCK (Pin 19).
 - RXD3 (Pin 22) is connected to RXD3 (Pin 22).
 - RXD2 (Pin 23) is connected to RXD2 (Pin 23).
 - RXD1 (Pin 24) is connected to RXD1 (Pin 24).
 - RXD0 (Pin 25) is connected to RXD0 (Pin 25).
 - RXDV (Pin 26) is connected to RXDV (Pin 26).
 - RXCK (Pin 27) is connected to RXCK (Pin 27).
- Data Signals:**
 - MDIO+ (Pin 1) is connected to MDIO+ (Pin 1).
 - MDIO- (Pin 2) is connected to MDIO- (Pin 2).
 - MDI+ (Pin 3) is connected to MDI+ (Pin 3).
 - MDI- (Pin 4) is connected to MDI- (Pin 4).
 - MDI2+ (Pin 5) is connected to MDI2+ (Pin 5).
 - MDI2- (Pin 6) is connected to MDI2- (Pin 6).
 - MDI3+ (Pin 7) is connected to MDI3+ (Pin 7).
 - MDI3- (Pin 8) is connected to MDI3- (Pin 8).
- Other Connections:**
 - PHY_RSTN (Pin 12) is connected to PHY_RSTN (Pin 12).
 - INTB/PMEB (Pin 31) is connected to INTB/PMEB (Pin 31).
 - CFG_EXT (Pin 32) is connected to CFG_EXT (Pin 32).
 - CFG_LD00 (Pin 33) is connected to CFG_LD00 (Pin 33).
 - CFG_LD01 (Pin 34) is connected to CFG_LD01 (Pin 34).
 - RSET (Pin 39) is connected to RSET (Pin 39).

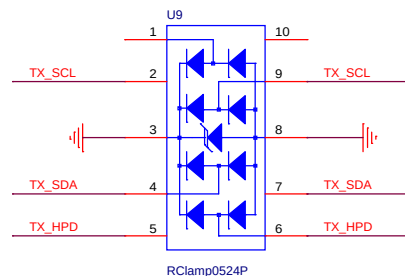
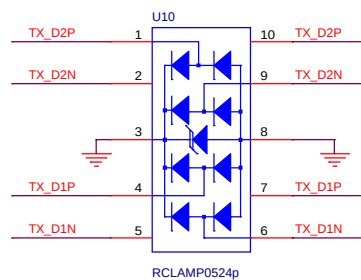
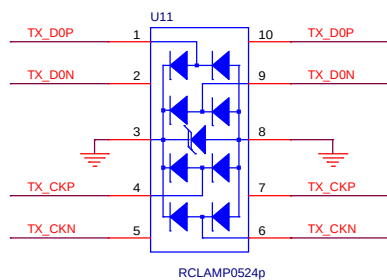
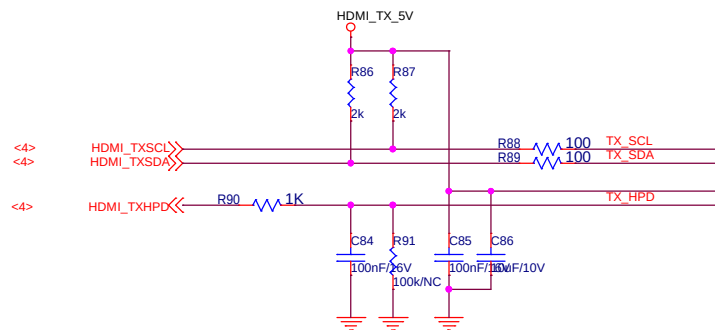
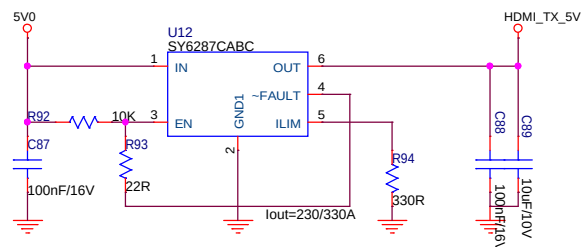
The RTL8211F-CG PHY is shown with its internal components, including resistors (R59, R61-R67, R68, R72, R73), capacitors (C78, C79, C67, C68, C69), a crystal (X1), and a pull-up resistor (R74). The PHY is connected to the microcontroller's pins via a series of resistors and capacitors.



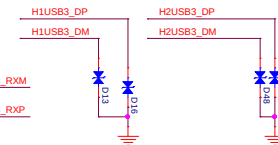
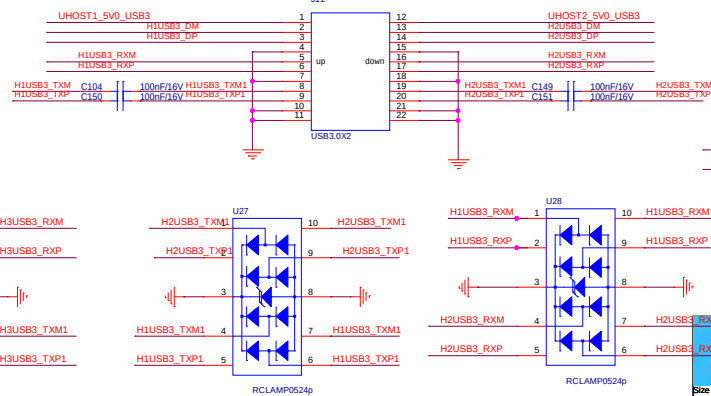
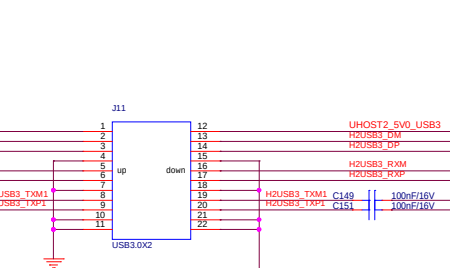
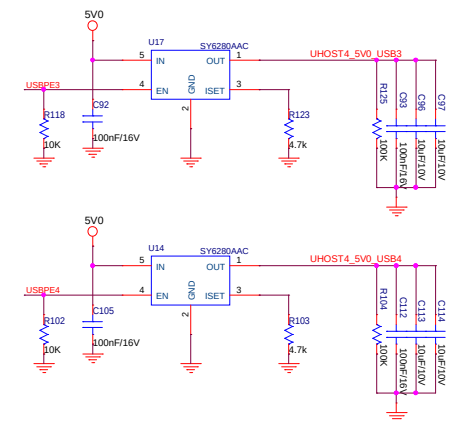
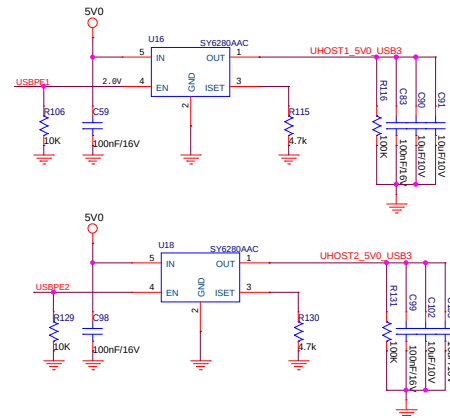
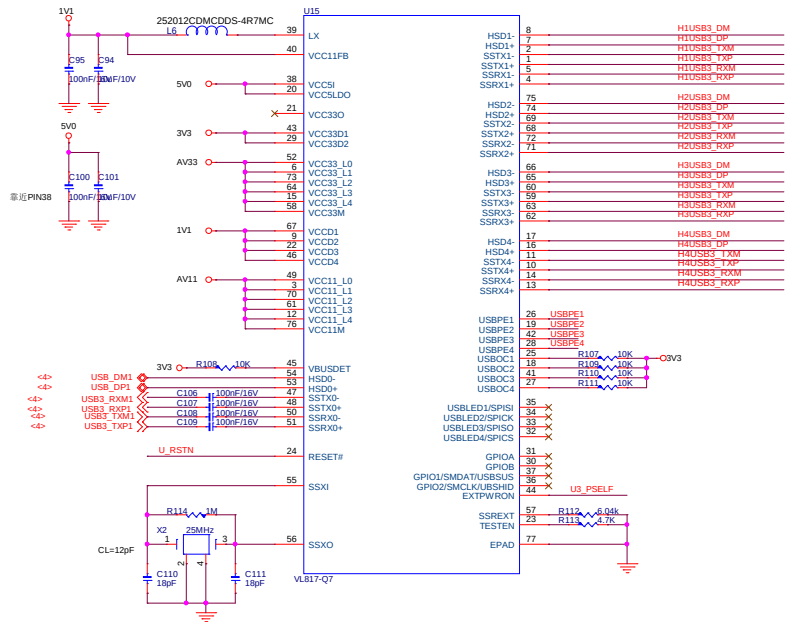
<4> HDMI_TX2P >> HDMI_TX2P TX_D2P
 <4> HDMI_TX2N >> HDMI_TX2N TX_D2N
 <4> HDMI_TX1P >> HDMI_TX1P TX_D1P
 <4> HDMI_TX1N >> HDMI_TX1N TX_D1N
 <4> HDMI_TX0P >> HDMI_TX0P TX_D0P
 <4> HDMI_TX0N >> HDMI_TX0N TX_D0N
 <4> HDMI_TXCP >> HDMI_TXCP TX_CKP
 <4> HDMI_TXCN >> HDMI_TXCN TX_CKN

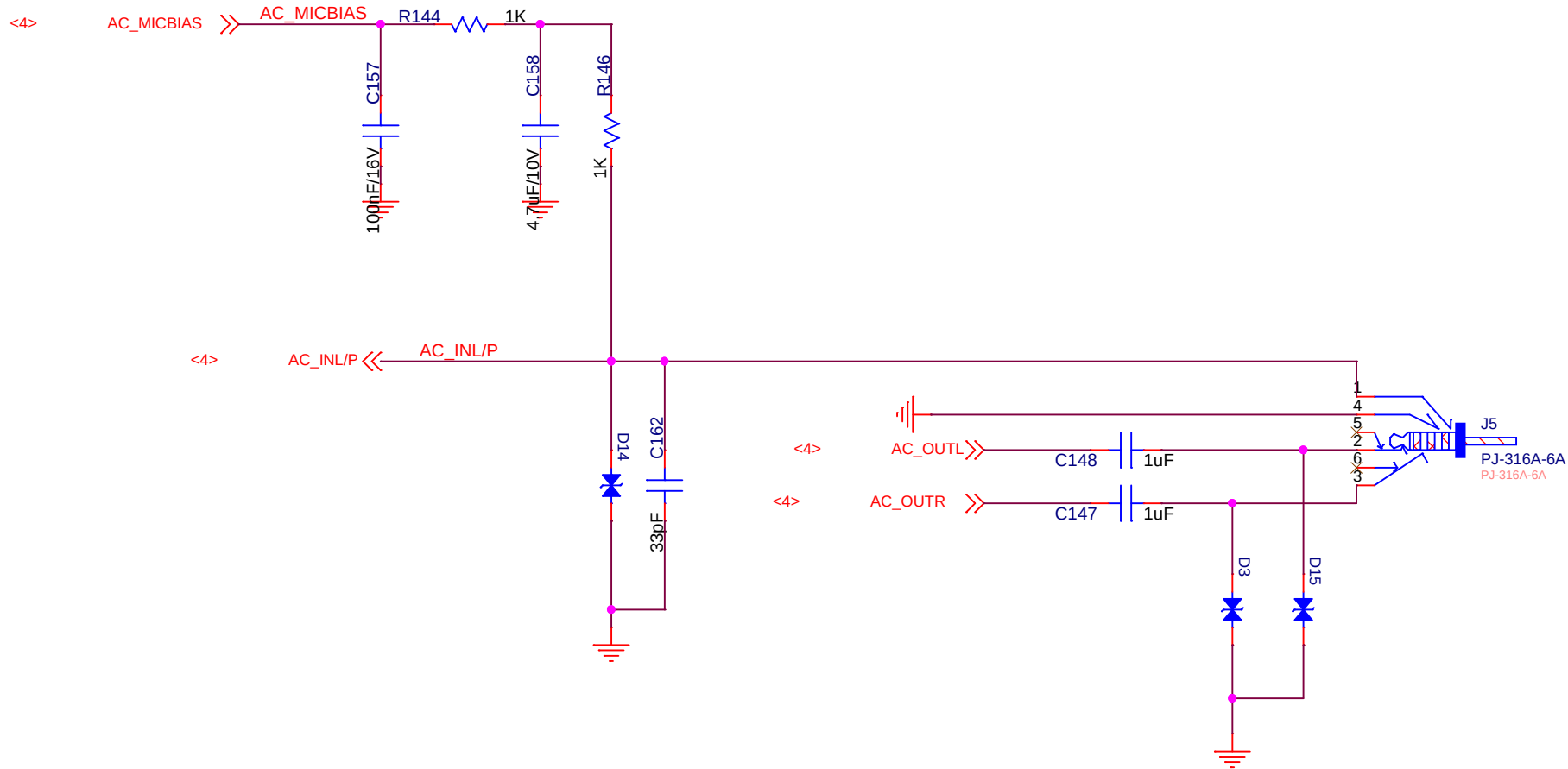


The HDMI differential trace impedance is 100ohm.



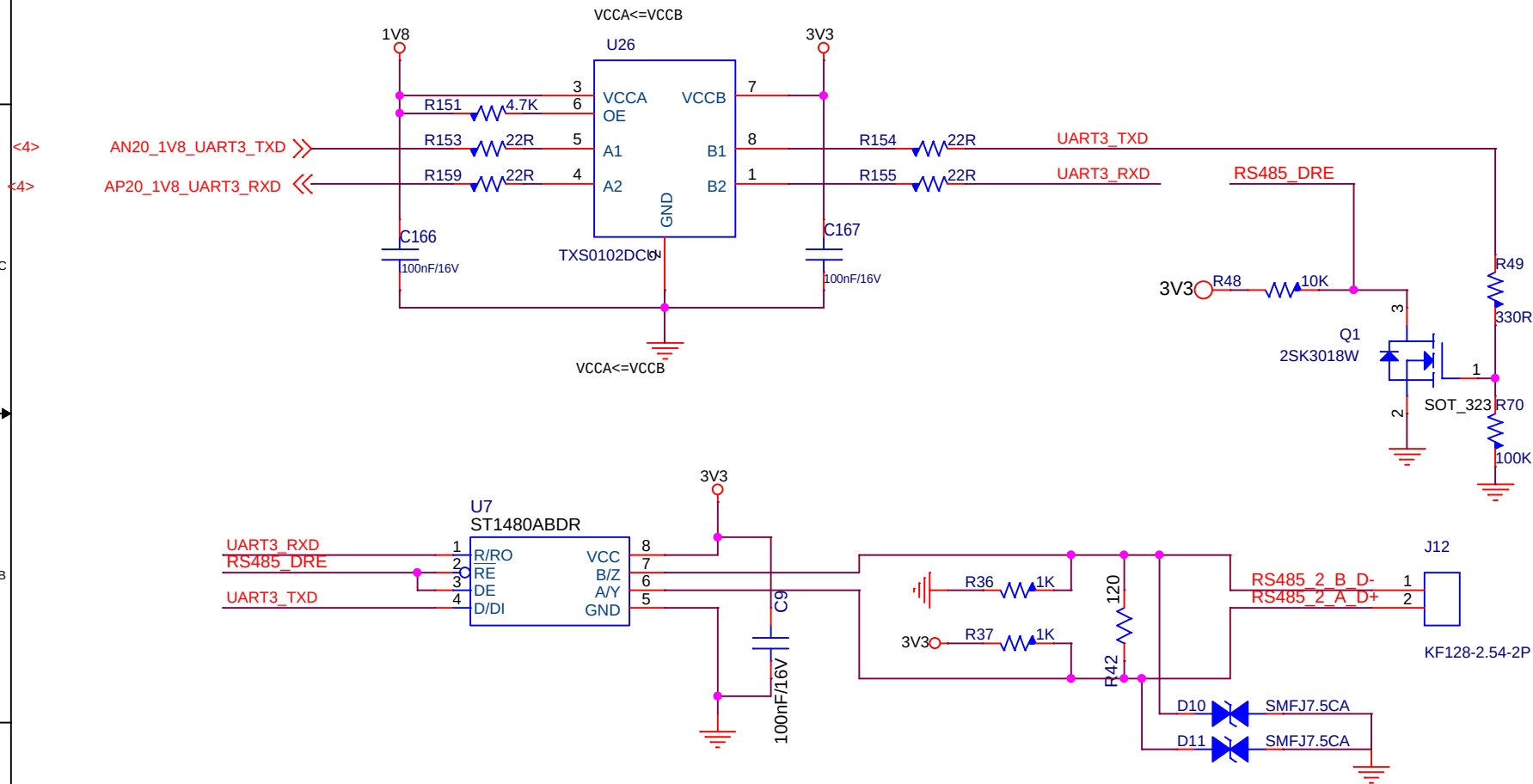
USB3.0





 Ebaina® 易百纳技术社区		WEBSITE:	
		http://bbs.ebaina.com	
Size A	DRAW: CHECK:	Roc.Zhou	Title <Title>
Date:	Monday, November 13, 2023	Sheet	9 of 17
		Rev	V0.01

RS485



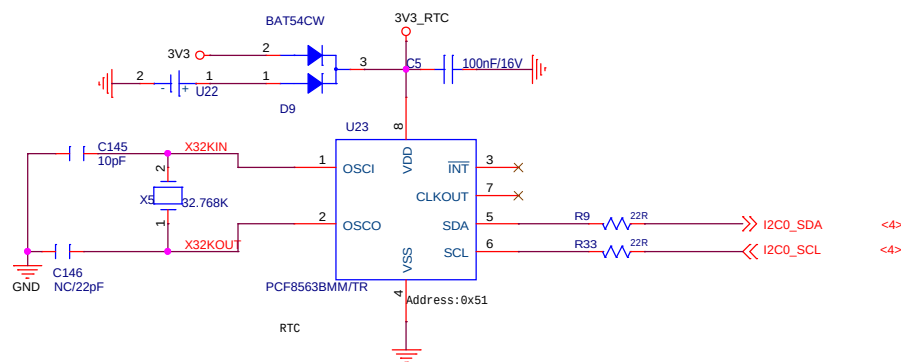
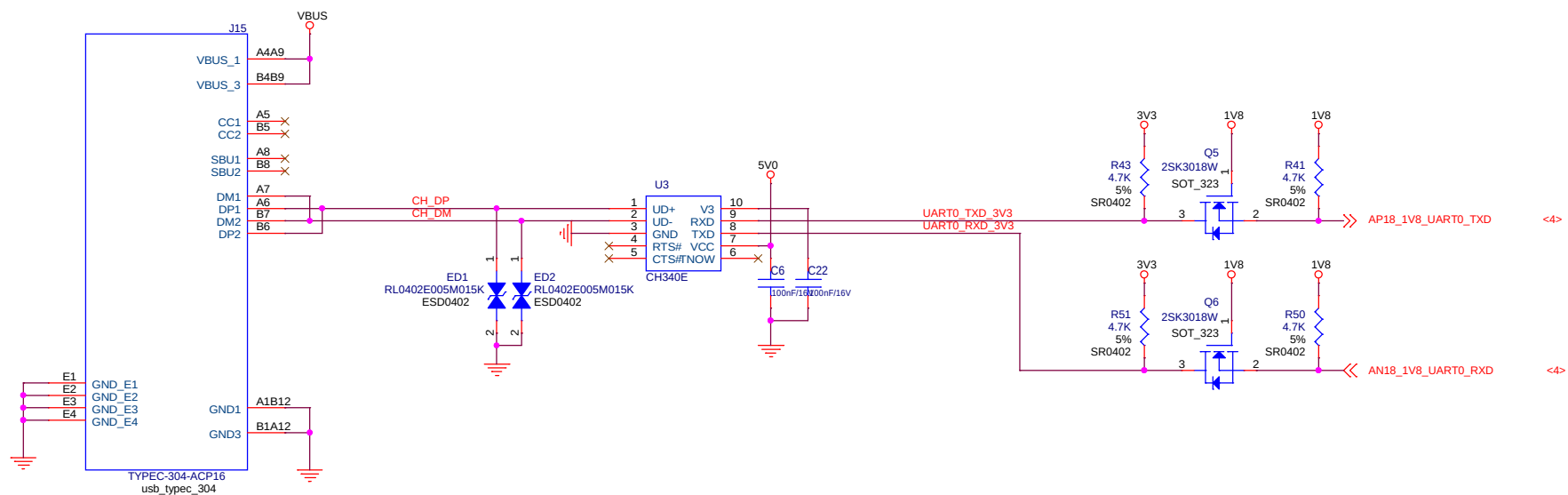


Ebaina
易百纳技术社区

WEBSITE:
<http://bbs.ebaina.com>

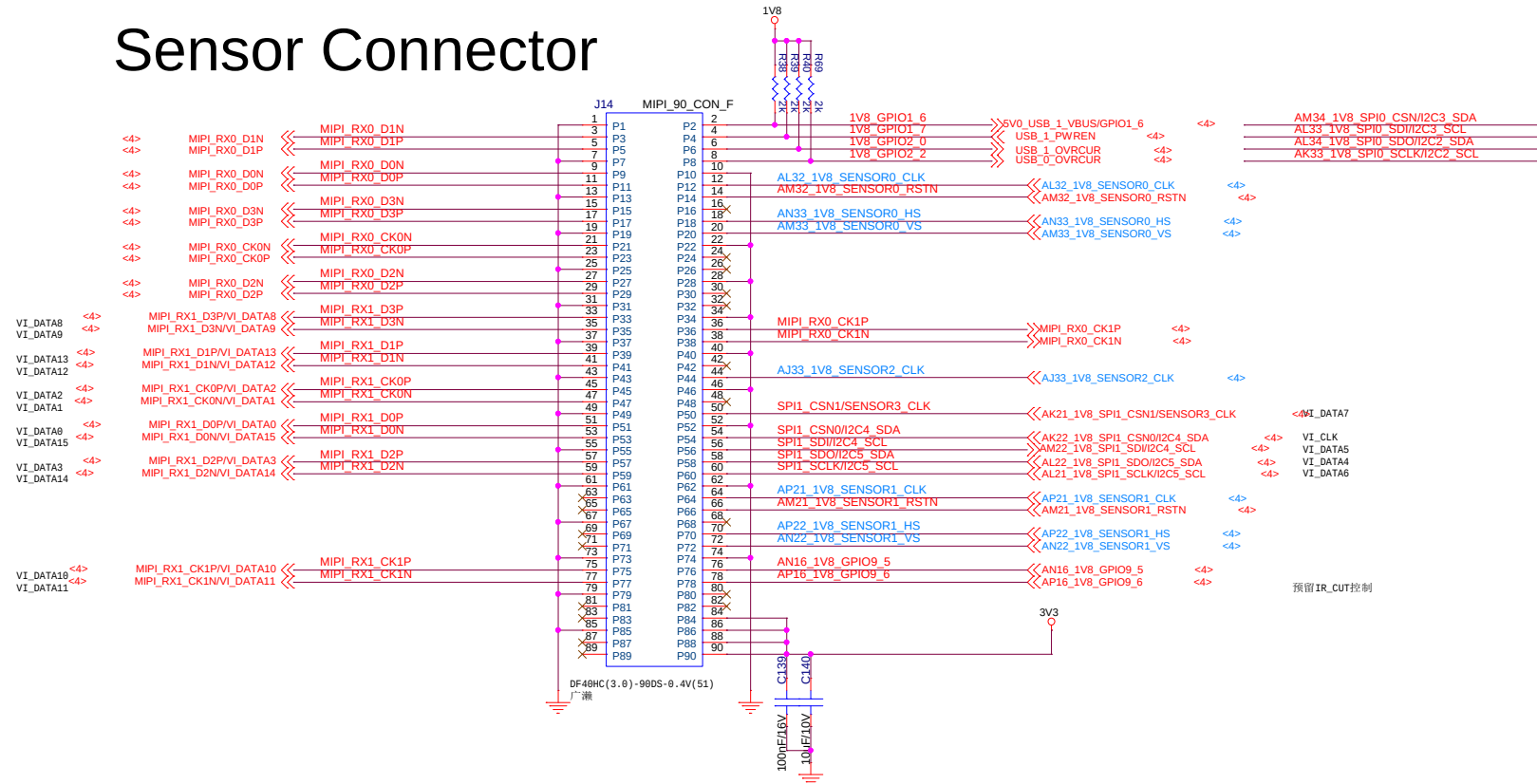
Size A	DRAW: CHECK:	Roc.Zhou	Title <Title>	Rev V0.01
Date: Friday, March 01, 2024		Sheet 10 of 17		

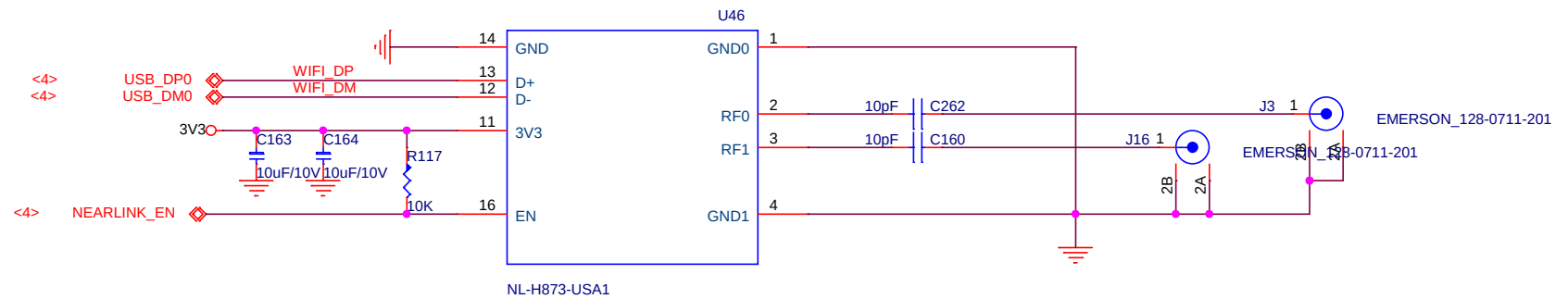
DEBUG

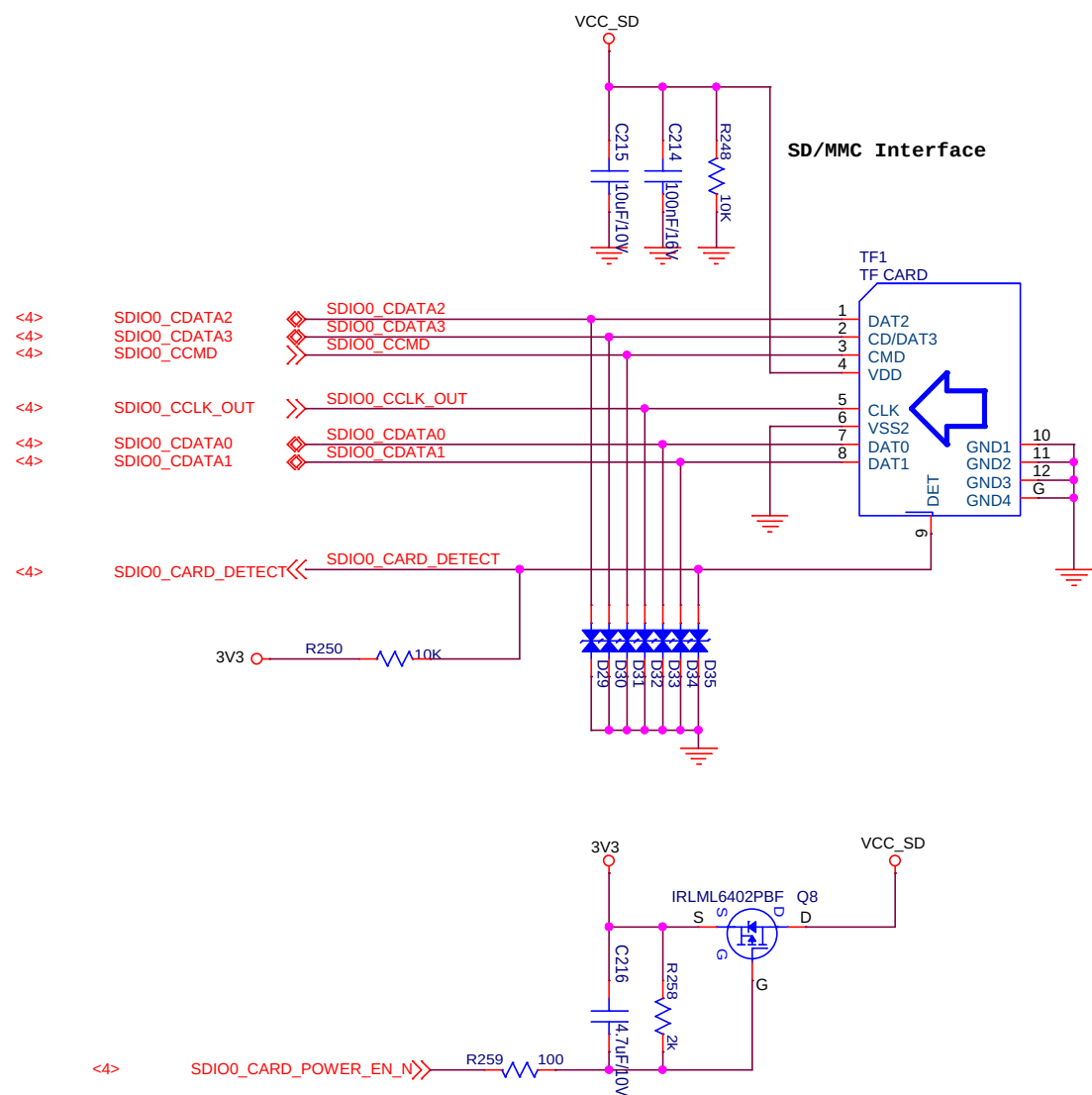


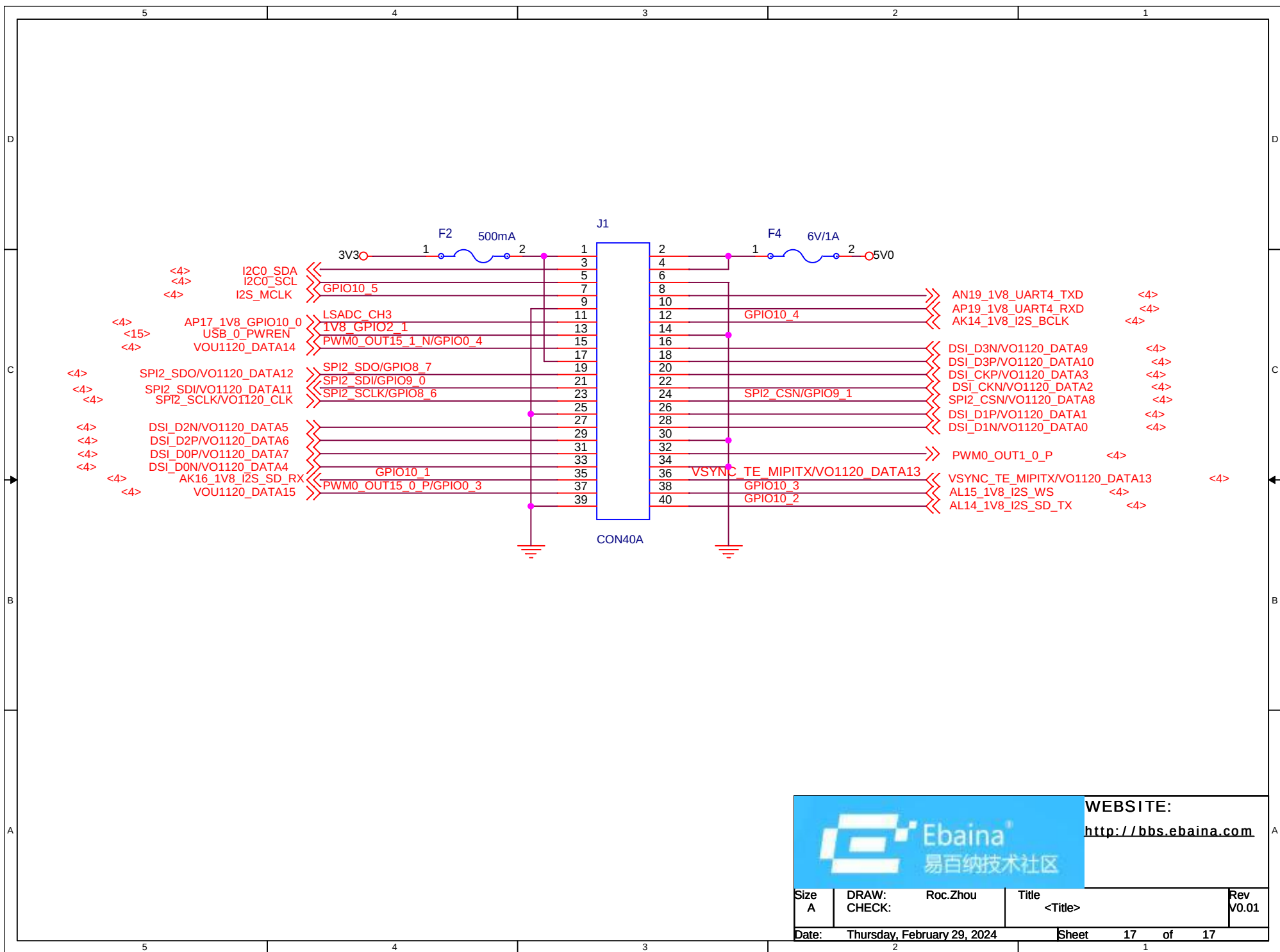
mipi_in

Sensor Connector











Ebaina
易百纳技术社区

WEBSITE:
<http://bbs.ebaina.com>

Size A	DRAW: CHECK:	Roc.Zhou	Title <Title>	Rev V0.01
Date: Thursday, February 29, 2024		Sheet 17 of 17		